Bayesian estimation II

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Plan for this slide deck

1. Bayesian forecasting
2. Modeling time-varying volatility
3. Bayesian VAR
4. Dealing with Covid-era data

Bayesian forecasting

- Recall the Bayesian AR(p) model from the previous slide deck.

$$y_t = c + a_1 y_{t-1} + \cdots + a_p y_{t-p} + u_t, \quad u_t \overset{iid}{\sim} N(0, \sigma^2).$$

- We've seen how to generate draws from the posterior $f(\boldsymbol{\theta} \mid \boldsymbol{Y})$ of the parameters $\boldsymbol{\theta} \equiv (c, a_1, \ldots, a_p, \sigma)'$ given data $\boldsymbol{Y} \equiv (y_1, y_2, \ldots, y_T)'$.
- How can we forecast y_{T+1}?
- More specifically, we want to characterize the one-step-ahead **predictive distribution** of y_{T+1} given $\boldsymbol{Y}$.

Bayesian forecasting (cont.)

- Density of predictive distribution:

$$f(y_{T+1} \mid \boldsymbol{Y}) = \int f(y_{T+1} \mid \boldsymbol{Y}, \boldsymbol{\theta}) f(\boldsymbol{\theta} \mid \boldsymbol{Y})\, d\boldsymbol{\theta}.$$

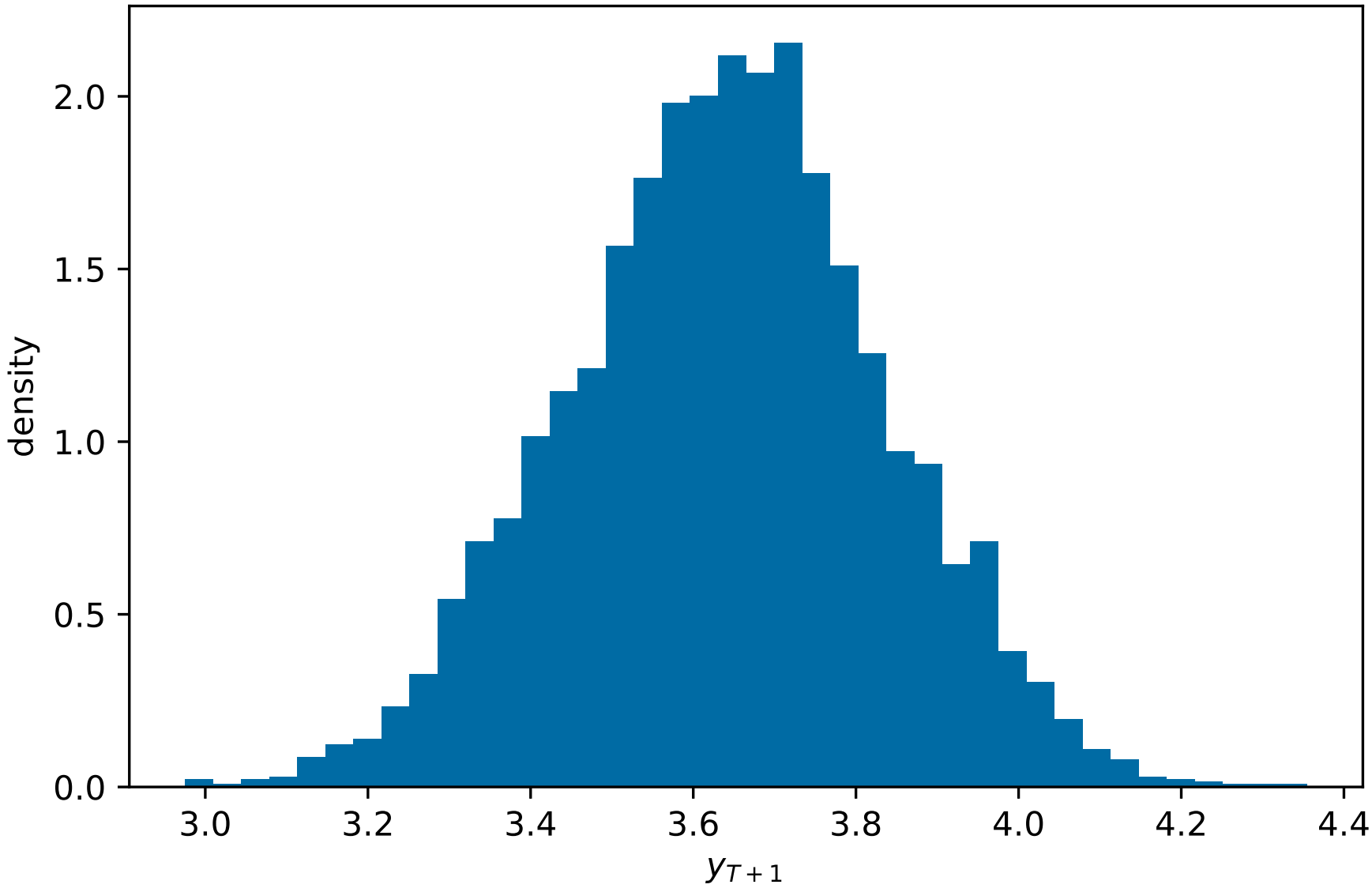
- Algorithm for simulating from the predictive distribution: for *each* posterior draw of $\boldsymbol{\theta}$, generate a draw y_{T+1} from

$$(y_{T+1} \mid \boldsymbol{Y}, \boldsymbol{\theta}) \sim N(c + a_1 y_T + \cdots + a_p y_{T-p+1}, \sigma^2).$$

- We can plot a histogram of the resulting draws $\{y_{T+1}\}$ to visualize the predictive distribution.
- The average draw approximates $E[y_{T+1} \mid \boldsymbol{Y}]$. Could also report percentiles to get a forecast interval.
- By design, the predictive distribution takes into account posterior uncertainty about the parameters $\boldsymbol{\theta}$.

```
# Extract parameter draws
draws = fit.draws_pd()
c_draws = draws['c'].values
a_draws = draws[[f'a[{l}]' for l in range(1, p+1)]].values
sigma_draws = draws['sigma'].values

# Simulate y_{T+1}
rng = np.random.default_rng(seed=31150)
x_new = y_full[-p:][::-1]  # p most recent observations in reverse order
mean_draws = c_draws + a_draws @ x_new # y_{T+1|T}
y_Tp1 = rng.normal(mean_draws, sigma_draws)
```


density
2.0
1.5
1.0
0.5
0.0
3.0
3.2
3.4
3.6
3.8
4.0
4.2
4.4
y_{T+1}

Time-varying volatility

- So far our Bayesian model has assumed i.i.d. shocks u_t. But many time series are likely heteroskedastic.
- **Autoregressive conditional heteroskedasticity** (ARCH) model:

$$(u_t \mid y_{t-1}, y_{t-2}, \dots) \sim N(0, \sigma_t^2),$$

$$\text{where} \quad \sigma_t^2 \equiv \omega + \gamma u_{t-1}^2.$$

  - $\{u_t\}$ is a martingale difference sequence. Under stationarity, we have $E[\sigma_t^2] = \omega/(1-\gamma)$ (exercise: verify).
- The AR(p) model with ARCH shocks (realistically) implies...
  1. **Volatility clustering:** when u_{t-1}^2 is large, u_t^2 is also likely large.
  2. y_t is harder to forecast in some periods than others:

$$\text{Var}(y_t \mid y_{t-1}, y_{t-2}, \dots) = \sigma_t^2 \neq \text{constant}.$$

Time-varying volatility: likelihood

$$(u_t \mid y_{t-1}, y_{t-2}, \dots) \sim N(0, \sigma_t^2), \quad \sigma_t^2 = \omega + \gamma u_{t-1}^2$$

- Include ω and γ in parameter vector $\boldsymbol{\theta}$.
- Likelihood function:

$$f(\boldsymbol{Y} \mid \boldsymbol{\theta}) = \underbrace{f(y_1, \dots, y_p \mid \boldsymbol{\theta})}_{\text{ignore}} \prod_{t=p+1}^{T} \frac{1}{\sqrt{2\pi\sigma_t^2}} e^{-(y_t - c - \boldsymbol{x}_t'\beta)^2/(2\sigma_t^2)},$$

where

$$\sigma_t^2 = \omega + \gamma(\underbrace{y_{t-1} - c - \boldsymbol{x}_{t-1}'\beta}_{=u_{t-1}})^2.$$

Stan model file:

```
data {
  int<lower=1> T;              // effective sample size
  int<lower=1> p;              // lag order
  vector[T] y;                 // observed series
  matrix[T, p] X;              // lag matrix
  real<lower=0> lambda;        // Minnesota prior tightness
  real<lower=0> scale_c;       // prior scale for AR intercept
  real<lower=0> scale_omega;   // prior scale for ARCH intercept
}

parameters {
  real c;                      // AR intercept
  vector[p] a;                 // AR coefficients
  real<lower=0> omega;         // ARCH intercept
  real<lower=0,upper=1> gamma; // ARCH parameter
}
```

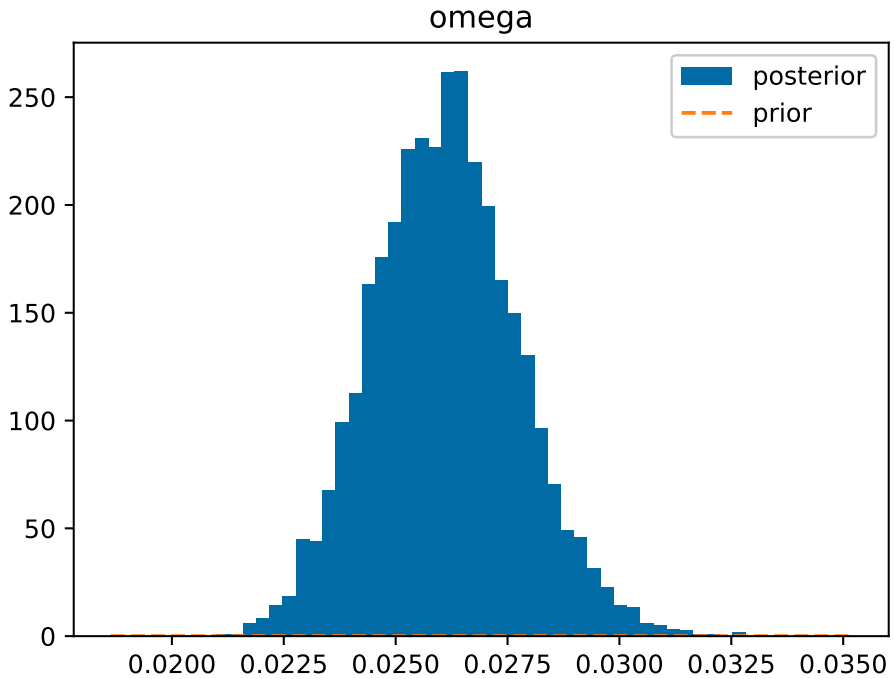
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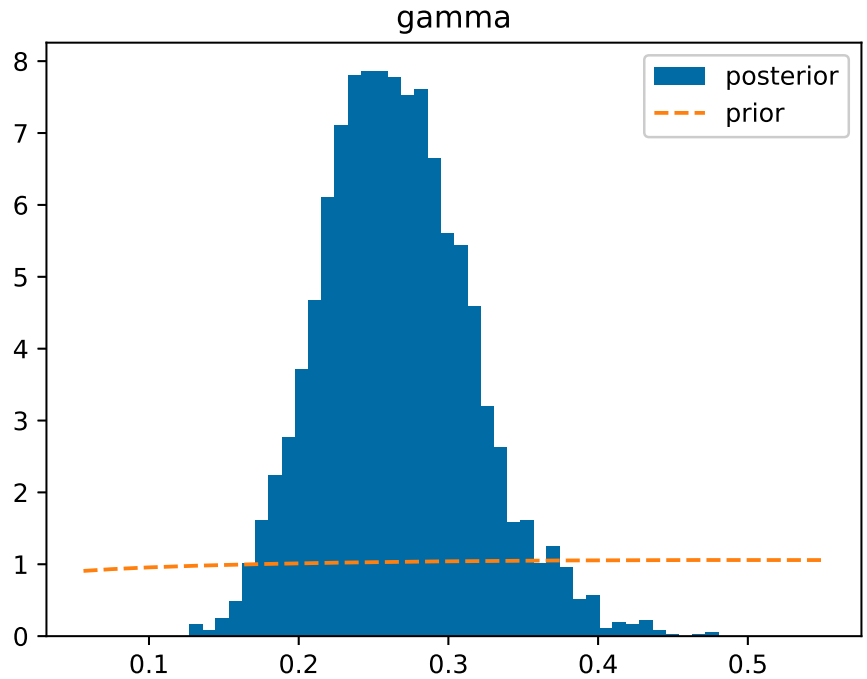
```
transformed parameters {
  vector[T] sigma2;            // conditional variances
  {
    vector[T] u = y - (c + X * a);
    sigma2[1] = omega / (1 - gamma); // initialize at unconditional mean
    for (t in 2:T)
      sigma2[t] = omega + gamma * square(u[t-1]);
  }
}

model {
  // Minnesota prior on AR coefficients
  for (l in 1:p) {
    real m_ = (l == 1) ? 1.0 : 0.0;
    real sd_ = lambda / l;
    a[l] ~ normal(m_, sd_);
  }

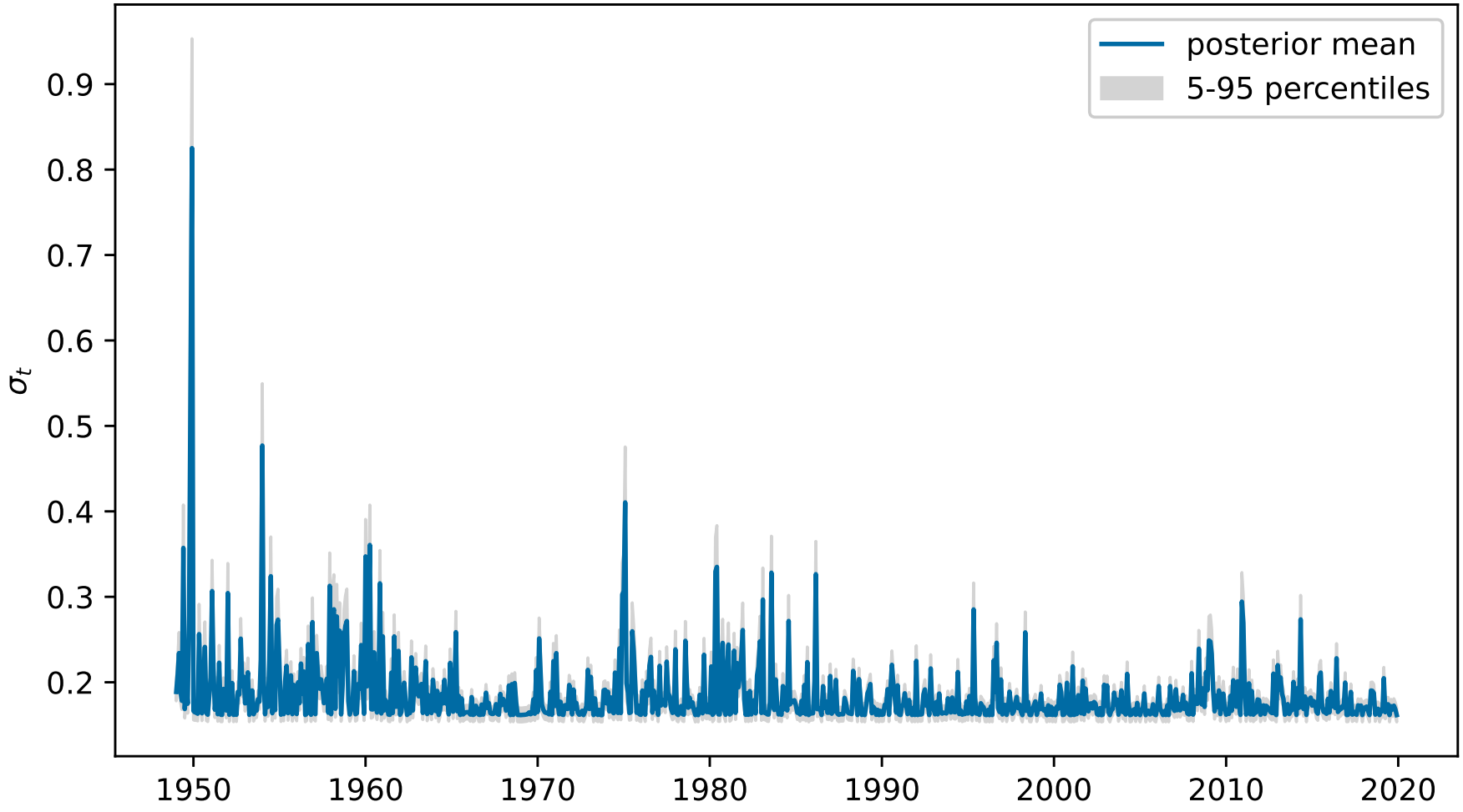
  // Intercept and ARCH priors
  c ~ normal(0, scale_c);
  omega ~ normal(0, scale_omega);
  gamma ~ beta(1.1, 1.1); // nearly uniform on [0,1]

  // Likelihood (vectorized)
  y ~ normal(c + X * a, sqrt(sigma2));
}
```

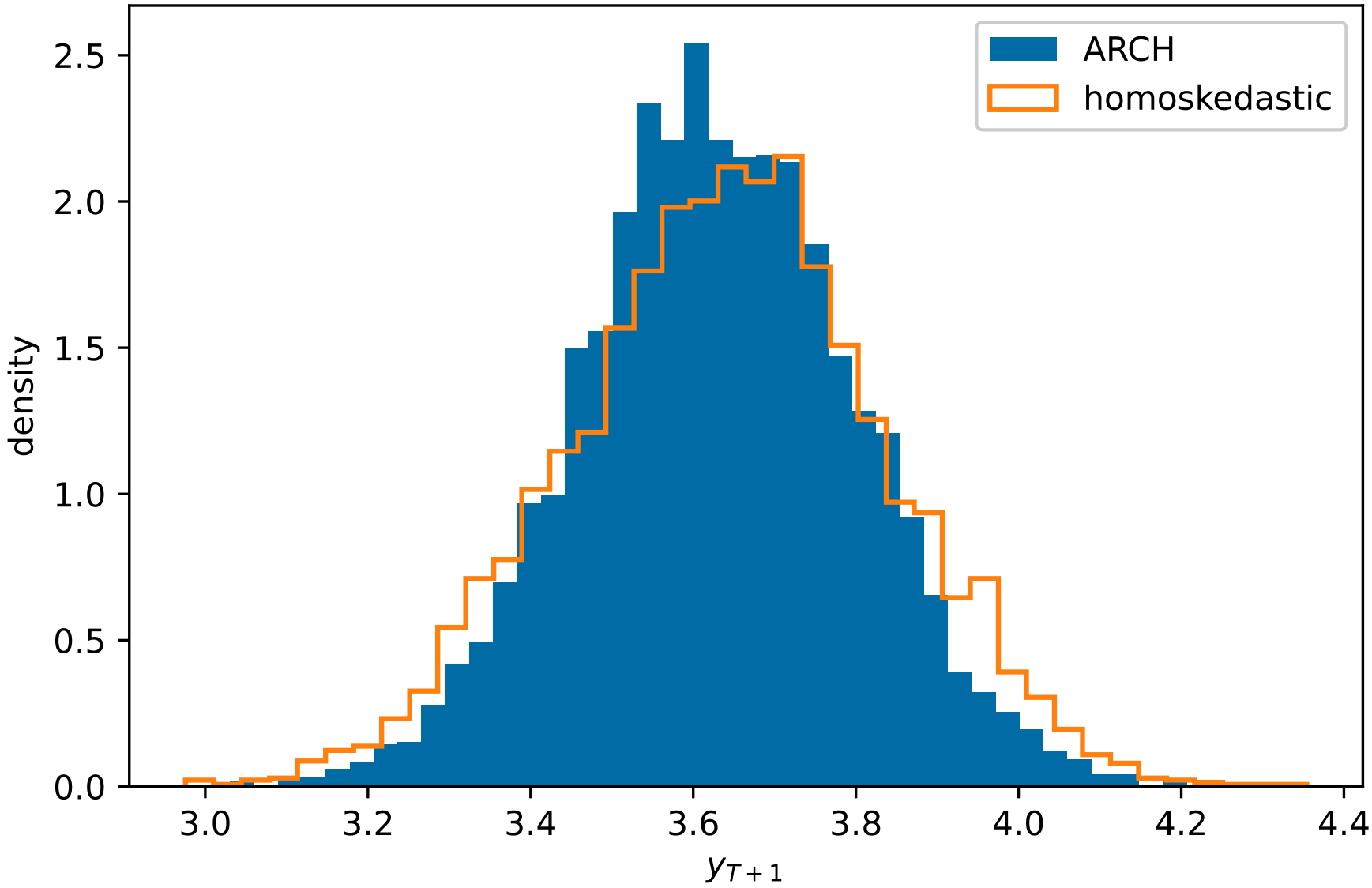




Note: prior on ω is so diffuse that it's barely visible.



Posterior distribution of $\{\sigma_t\}_{t=p+1}^{T}$ – notice volatility clustering



One-step-ahead predictive distribution: ARCH vs. homoskedastic shocks

Time-varying volatility: GARCH

- **Generalized ARCH** (GARCH) model:

$$(u_t \mid y_{t-1}, y_{t-2}, \dots) \sim N(0, \sigma_t^2),$$

$$\sigma_t^2 = \omega + \gamma u_{t-1}^2 + \delta \sigma_{t-1}^2.$$

- Could include more lags of u_t^2 and/or σ_t^2 on the right-hand side.
- You'll be working with this model on the problem set.

Bayesian VAR (BVAR)

- VAR(p) model (homoskedastic for simplicity):

$$\boldsymbol{y}_t = \boldsymbol{c} + \boldsymbol{A}_1 \boldsymbol{y}_{t-1} + \cdots + \boldsymbol{A}_p \boldsymbol{y}_{t-p} + \boldsymbol{u}_t, \quad \boldsymbol{u}_t \overset{iid}{\sim} N(\boldsymbol{0}, \boldsymbol{\Sigma}).$$

- Data: $\boldsymbol{Y} \equiv (\boldsymbol{y}_1, \ldots, \boldsymbol{y}_T)'$, dimension $T \times n$.
- Parameters: $\boldsymbol{\theta} = \{\boldsymbol{c}, \boldsymbol{A}_1, \ldots, \boldsymbol{A}_p, \boldsymbol{\Sigma}\}$.

BVAR: Minnesota prior

- $A_{\ell,ij}$: (i,j) element of $\boldsymbol{A}_\ell$.
- Prior means:

$$E[A_{\ell,ij}] = \begin{cases} 1 & \text{if } i = j \text{ and } \ell = 1, \\ 0 & \text{otherwise.} \end{cases}$$

- Each variable y_{it} is shrunk towards a univariate random walk: $y_{it} = y_{i,t-1} + u_{it}$.
- Prior belief: limited Granger causality from one variable to another.

BVAR: Minnesota prior (cont.)

- Prior variances:

$$\text{Var}(A_{\ell,ij}) = \begin{cases} \left(\frac{\lambda}{\ell}\right)^2 & \text{if } i = j, \\ \left(\frac{\lambda\kappa}{\ell} \times \frac{\hat{\sigma}_i}{\hat{\sigma}_j}\right)^2 & \text{otherwise.} \end{cases}$$

- λ controls overall prior variance.
- κ controls relative variance on *cross-variable* coefficients ($i \neq j$).
  - Typical values in literature: $\lambda = 0.2$, $\kappa = 0.3$.
- Lower variance (more dogmatic beliefs) on longer lags ℓ.
- $\hat{\sigma}_i$: estimate of $\sqrt{\text{Var}(u_{it})}$ (e.g., from preliminary OLS VAR).
- Scaling $\hat{\sigma}_i/\hat{\sigma}_j$ adjusts for disparate units of y_{it} and y_{jt}.

BVAR: Minnesota prior (cont.)

- Example: bivariate VAR(2), with priors in blue.

$$y_{1t} = c_1 + \underset{N(1,\lambda^2)}{A_{1,11}}\, y_{1,t-1} + \underset{N\left(0,(\lambda\kappa\frac{\hat{\sigma}_1}{\hat{\sigma}_2})^2\right)}{A_{1,12}}\, y_{2,t-1} + \underset{N\left(0,(\frac{\lambda}{2})^2\right)}{A_{2,11}}\, y_{1,t-2} + \underset{N\left(0,(\frac{\lambda\kappa}{2}\frac{\hat{\sigma}_1}{\hat{\sigma}_2})^2\right)}{A_{2,12}}\, y_{2,t-2} + u_{1t}$$

$$y_{2t} = c_2 + \underset{N\left(0,(\lambda\kappa\frac{\hat{\sigma}_2}{\hat{\sigma}_1})^2\right)}{A_{1,21}}\, y_{1,t-1} + \underset{N(1,\lambda^2)}{A_{1,22}}\, y_{2,t-1} + \underset{N\left(0,(\frac{\lambda\kappa}{2}\frac{\hat{\sigma}_2}{\hat{\sigma}_1})^2\right)}{A_{2,21}}\, y_{1,t-2} + \underset{N\left(0,(\frac{\lambda}{2})^2\right)}{A_{2,22}}\, y_{2,t-2} + u_{2t}$$

Excerpts from Stan model file:

```
transformed data {
  // Build mean and stdev of Minnesota prior
  matrix[n, n*p] A_mean;
  matrix[n, n*p] A_std;
  for (l in 1:p) {      // lag
    for (i in 1:n) {    // equation
      for (j in 1:n) {  // regressor
        int col = (l - 1) * n + j;
        A_mean[i, col] = (l == 1 && i == j) ? 1 : 0;
        A_std[i, col] = (i == j)
          ? lambda / l
          : (lambda / l) * (kappa * scale_sigma[i] / scale_sigma[j]);
      }
    }
  }

  // Scale matrix for Sigma prior
  matrix[n, n] Sigma_scale = diag_matrix(square(scale_sigma));
}
```

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```
model {
  // Minnesota prior on VAR coefficients
  to_vector(A) ~ normal(to_vector(A_mean), to_vector(A_std));

  // Intercept prior
  c ~ normal(0, scale_c);

  // Prior on innovation var-cov matrix
  Sigma ~ inv_wishart(n + 2, Sigma_scale);

  // Likelihood
  for (t in 1:T)
    Y[t] ~ multi_normal(c + A * X[t]', Sigma);
}
```

Note: the inverse-Wishart distribution is a distribution on the space of $n \times n$ variance-covariance matrices.

BVAR: sampling

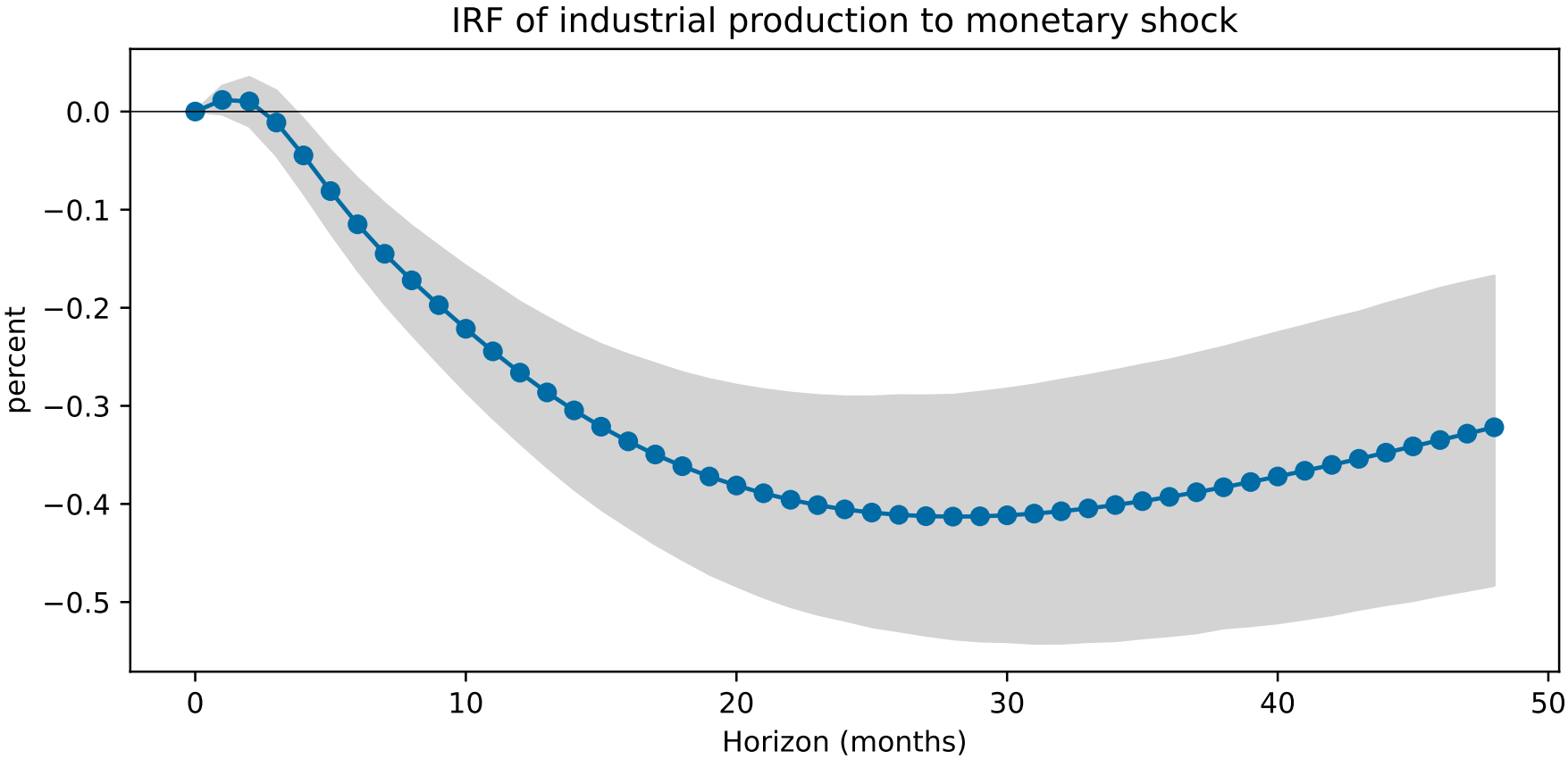
- Stan takes substantially longer to sample from this model due to the many parameters.
- One can design much faster samplers for this specific model and prior. There is a huge literature on computational tricks for BVARs.
- Stan, however, makes it easy to modify the model or prior (e.g., add time-varying volatility).

Bayesian SVAR

- How to use BVARs for causal inference?
- Given the posterior draws of the VAR parameters $\boldsymbol{\theta} = \{\boldsymbol{c}, \boldsymbol{A}_1, \ldots, \boldsymbol{A}_p, \boldsymbol{\Sigma}\}$, we can compute the Cholesky-identified IRFs corresponding to each draw:

$$\boldsymbol{\Theta}_h = \left(\boldsymbol{I}_{n\times n}, \boldsymbol{0}_{n\times n(p-1)}\right) \boldsymbol{A}^h \begin{pmatrix} \boldsymbol{H} \\ \boldsymbol{0}_{n(p-1)\times n} \end{pmatrix},$$

  where $\boldsymbol{H} \equiv \text{Cholesky}(\boldsymbol{\Sigma})$, and $\boldsymbol{A}$ is the $(np) \times (np)$ companion form matrix corresponding to $\boldsymbol{A}_1, \ldots, \boldsymbol{A}_p$.
- This gives us posterior draws of the structural IRFs.
- Similarly for FEVDs.



6-variable Cholesky-identified SVAR (as in earlier slide deck)

Blue dots: posterior means; gray band: posterior 68% credible intervals

Hierarchical Bayes

- Results may be sensitive to the **hyper-parameters** λ and κ in the Minnesota prior, since they determine the degree of shrinkage.
- It would be nice to have a data-dependent way to select them.
- **Hierarchical Bayes**: treat these as unknown parameters themselves, with their own "hyper-priors".
- In Stan, just move `lambda` and `kappa` to the `parameters` block. In the `model` block, add their priors:

```
lambda ~ gamma(0.25, 1.25); // mean 0.2, stdev 0.4
kappa ~ gamma(0.25, 0.833); // mean 0.3, stdev 0.6
```

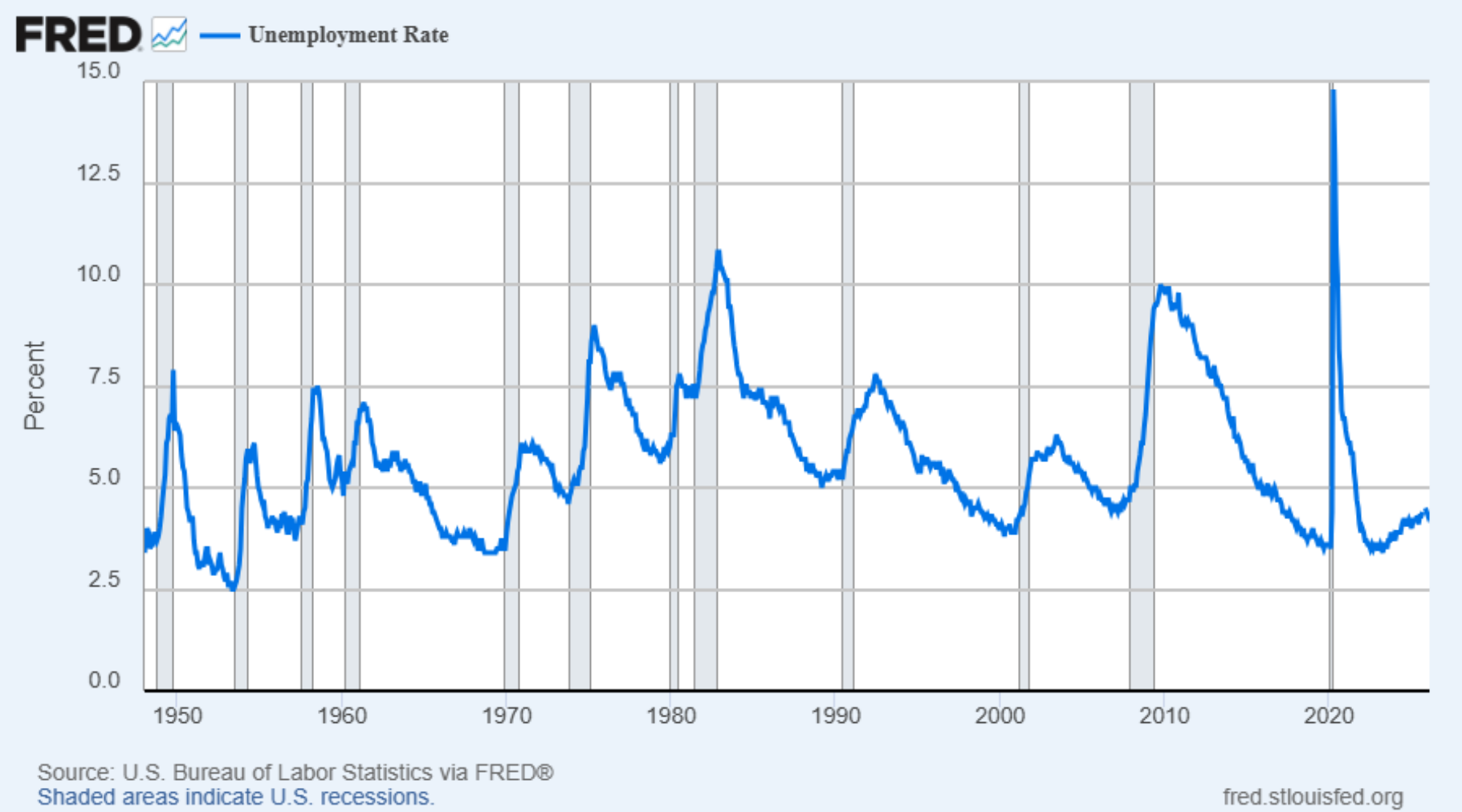
- Further reading: Giannone, Lenza & Primiceri (2015).

Prior shrinkage and causal inference

- Prior shrinkage reduces the variance of estimators, but at the expense of higher bias (unless our prior beliefs are totally accurate).
- BVARs often outperform OLS VAR estimates when it comes to mean squared error (for forecasts or parameters).
- However, as we discussed when comparing LP and SVAR, even a bit of bias can cause severe coverage distortions. These distortions get worse the tighter are our priors.
- I would not recommend using BVARs with the conventional Minnesota prior for *causal inference*. BVARs should only be employed with a very long lag order p and much more diffuse priors than we have imposed above.

Modeling the Covid shock

- Textbook time series models have a difficult time dealing with data from March 2020 and the following few months.
- Key economic time series showed extreme jumps/swings in these months.
- If we estimate a model that does not explicitly model the Covid period, our estimated shock volatility parameters will be largely driven by those few hard-to-fit observations.
  - Could mess up causal inference or post-2020 forecasts.


FRED
Unemployment Rate
15.0
12.5
10.0
7.5
5.0
2.5
0.0
Percent
1950
1960
1970
1980
1990
2000
2010
2020
Source: U.S. Bureau of Labor Statistics via FRED®
Shaded areas indicate U.S. recessions.
fred.stlouisfed.org

Modeling the Covid shock (cont.)

- We can allow for heightened volatility in the Covid period by letting our shock volatility parameter shift in those periods.
- AR(p) example:

$$y_t = c + a_1 y_{t-1} + \cdots + a_p y_{t-p} + u_t, \quad u_t \overset{indep}{\sim} N(0, \sigma_t^2),$$

$$\text{where} \quad \sigma_t \equiv (1 + \eta s_t)\sigma, \quad s_t \equiv \begin{cases} 1 & \text{for } t \in \text{Covid}, \\ 0 & \text{otherwise.} \end{cases}$$

- The parameter $\eta \geq 0$ captures the volatility shift.
- Estimation in Stan only requires minor modifications of our earlier AR(p) model file.
- More detailed guidance: Lenza & Primiceri (2022).

What did we learn?

- Bayesian forecasting, predictive distribution
- ARCH, GARCH
- BVAR
  - Minnesota prior
  - Cholesky identification
  - Hierarchical Bayes
- Modeling Covid as a volatility shift

Looking ahead

- State space models: very rich class of models with latent variables
- Inferring the values of latent variables
- Factor models